

Deepika Manne

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M.S. Mathematical Finance candidate at Boston College with 3+ years of software engineering experience and hands-on experience in quantitative research, derivatives pricing, systematic trading, portfolio analytics, and financial markets. Strong foundation in Python, statistics, financial modeling, portfolio construction, risk analytics, and execution-aware backtesting, with experience building scalable analytical systems for institutional datasets.

EDUCATION

Boston College, Carroll School of Management.

Chestnut Hill, MA | Expected Dec 2026

Master of Science in Mathematical Finance | Dean's Scholarship | GPA: 3.68

Jawaharlal Nehru Technological University.

Hyderabad, India | Jun 2018

Bachelor of Technology in Electronics and Communication Engineering | Full Merit Scholarship.

TECHNICAL & QUANTITATIVE SKILLS

Programming: Python, SQL, C++, R, Java.

Quantitative: Statistics, Econometrics, Factor Modeling, Portfolio Optimization, Backtesting, Risk Analysis, Systematic Trading.

Data & Research: Polars, pandas, NumPy, SciPy, statsmodels, scikit-learn, CVXPY.

Tools: PostgreSQL, Git, AWS, Excel, Bloomberg.

Certificates and Credentials: Bloomberg Market Concepts, Bloomberg Finance Fundamentals, FINRA Securities Industry Essentials (SIE).

EXPERIENCE

Full Stack Developer

October 2023 – November 2025

Intrafi — Toronto, Canada

- Built portfolio analytics and data applications using Python, React, PostgreSQL, and API-driven workflows, improving dashboard performance by 40% through optimized data processing and application architecture.
- Engineered automated pipelines for institutional portfolio data and performance metrics, eliminating 15+ hours of manual work per week and improving reporting reliability and timeliness.

Front End Developer

June 2022 – September 2023

Sabio Inc. — Toronto, Canada

- Developed data-intensive Angular applications and analytical dashboards for large institutional datasets, translating complex business requirements into scalable production systems and interactive analytical workflows.

PROJECTS

Institutional Quantitative Research Platform

Python | Polars | DuckDB | pandas | NumPy | SciPy | statsmodels | CVXPY | pytest

- Built a modular systematic-research platform for market and fundamental data ingestion, feature engineering, factor research, portfolio construction, backtesting, risk analytics, and automated reporting, designed to support large U.S. equity research universes.
- Researched momentum, mean-reversion, low-volatility, liquidity, value, and quality factors using IC, Rank IC, ICIR, quantile spreads, factor decay, turnover, cross-factor correlations, and False Discovery Rate controls, with point-in-time and historical-universe safeguards against look-ahead and survivorship bias.
- Developed long-only and dollar-neutral portfolios using inverse-volatility, minimum-variance, mean-variance, and maximum-Sharpe-style methods with position, sector, beta, turnover, and liquidity constraints.
- Engineered walk-forward, execution-aware backtesting incorporating transaction costs, bid-ask spreads, slippage, liquidity limits, Sharpe, drawdown, VaR/CVaR, tracking error, turnover, and attribution; validated the end-to-end workflow on a controlled equity sample with 33 passing pytest tests.

WTI Crude Oil Return & Volatility Modeling | Academic Research Project

Python | pandas | NumPy | SciPy | statsmodels | statsmodels

- Constructed a forecasting dataset of 422 monthly observations from 1990–2026 with 17 lagged financial, energy, macroeconomic, and geopolitical predictors, applying stationarity transformations and forecasting-safe lag structures to reduce look-ahead risk.
- Estimated HC3-robust OLS models, finding greater predictability in volatility (adjusted $R^2 = 0.230$; $p = .022$) than directional returns (adjusted $R^2 = 0.071$; $p = .138$); diagnostics identified significant volatility clustering (ARCH-LM = 54.69; $p < .001$) and fat-tailed residuals.
- Designed follow-on comparisons of Gaussian GARCH, Student-t GARCH, GJR-GARCH, and GARCH-X evaluation with rolling out-of-sample testing using QLIKE and Diebold–Mariano test against benchmark forecasts.

Derivatives Pricing & Risk Analytics Engine

Python | Options Pricing | Black–Scholes | Monte Carlo | Greeks | Volatility | Hedging

- Built European/American option pricing and risk analytics using Black–Scholes, CRR, Monte Carlo, implied-volatility surfaces, portfolio Greeks, and scenario P&L with convergence and cross-model validation.
- Engineered a dynamic delta-hedging simulator incorporating realized volatility, rebalancing frequency, transaction costs, and hedging P&L to analyze replication performance and model risk.

Macro FX Trading Simulation Project

EUR/USD | USD/JPY | Macroeconomic Analysis | Trade Execution | Risk Management

- Managed a simulated EUR/USD book across 37 trades and €218.3M gross turnover, generating approximately \$1.25M in platform-reported simulated P&L while forming catalyst-driven views from labor data, FOMC guidance, and FX-intervention developments within a €100M position limit.